Practical 8

Aim : Study about Hyperledger Fabric.

Introduction :

Hyperledger Fabric is an open-source, enterprise-grade permissioned blockchain platform developed under the Hyperledger project. It is designed for organizations that need a secure blockchain network where participants are known and authorized.

Unlike public blockchains such as Bitcoin and Ethereum, Hyperledger Fabric allows organizations to control who can join the network and what they are allowed to access or perform.

Fabric uses smart contracts, known as chaincode, to implement business logic and record transactions on a distributed ledger.

Main Components :

1. Organizations

Organizations represent the different companies or entities participating in the blockchain network.

Examples: Bank, Supplier, Manufacturer, Government Organization.

2. Peers

Peers are network nodes that maintain the ledger and execute transactions. They can also endorse transactions according to the rules of the network.

3. Ordering Service

The ordering service receives transactions from the network, puts them in the correct order, and distributes them to the appropriate peers.

4. Ledger

The ledger stores the transaction history of the blockchain network. It consists of the blockchain itself and the current state of the network.

5. Chaincode

Chaincode is the **smart contract** of Hyperledger Fabric. It contains the business rules and defines how transactions can change the state of the ledger.

6. Channels

Channels provide a private communication and transaction environment between selected organizations. They allow data to be shared only with authorized participants.

7. Certificate Authority

The Certificate Authority (CA) is responsible for providing digital identities and certificates to participants in the network.

Basic Architecture :

The basic working structure of Hyperledger Fabric can be represented as:

Client Application

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Peer Nodes

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Chaincode

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Ordering Service

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Peer Nodes

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Distributed Ledger

Working of Hyperledger Fabric

1. A client application submits a transaction request.
2. The transaction is sent to the required peer nodes for endorsement.
3. The chaincode checks and executes the transaction.
4. The endorsed transaction is sent to the ordering service.
5. The ordering service places transactions in order and creates blocks.
6. The blocks are distributed to peer nodes.
7. Peers validate the transactions and update their copy of the ledger.

Features

- Permissioned blockchain network.
- Provides identity management.
- Supports smart contracts through chaincode.

- Provides channels for private transactions.
- Supports multiple organizations.
- Provides a distributed and tamper-resistant ledger.
- Suitable for enterprise applications.

Advantages

- Provides better privacy and access control.
- Suitable for business and enterprise applications.
- Known participants improve network governance.
- Supports modular and flexible architecture.
- Can handle different business requirements.

Applications

Hyperledger Fabric can be used in:

- Supply chain management
- Banking and finance
- Healthcare
- Insurance
- Digital identity
- Trade and logistics
- Asset tracking
- Government services

Conclusion

Thus, **Hyperledger Fabric**, its architecture, components, working, features, advantages, and applications were studied and understood. It provides a secure and permissioned blockchain platform suitable for developing enterprise-level blockchain applications.